Westfield University

BSc Computer Science & Artificial Intelligence

Official Module Results · Year 2, Semester 1 · Academic Year 2024–25

Student:	Jordan Mercer	**Student ID:**	S4821903	**Semester GPA:**	**81/100 · B**
Programme:	BSc CS & AI	**Year / Sem:**	Year 2, Sem 1	**Advisor:**	Prof. Sarah Lin

MATH201 · Linear Algebra — 74% Grade: C

Vector Spaces — 76% · C

Basis & Dimension	72%	C
Linear Transformations	79%	C

Matrix Methods — 68% · D

Eigenvalues & Eigenvectors	61%	D
Matrix Decomposition	76%	C

MATH202 · Probability & Statistics — 77% Grade: C

Probability Theory — 78% · C

Bayes Theorem	77%	C
Distributions & Expectation	78%	C

Statistical Inference — 87% · B

Hypothesis Testing	89%	B
Max Likelihood Estimation	85%	B

CS201 · Algorithms & Data Structures — 83% Grade: B

Algorithm Design — 84% · B

Divide & Conquer	80%	B
Dynamic Programming	87%	B

Graph Algorithms — 84% · B

Shortest Paths	82%	B	■■■■■■■■■■■■□□□
Minimum Spanning Trees	86%	B	■■■■■■■■■■■■□□□

CS202 · Machine Learning — 77% Grade: C

Supervised Learning — 80% · B

Linear & Logistic Regression	78%	C	■■■■■■■■■■■□□□□
Decision Trees & Ensembles	81%	B	■■■■■■■■■■■■□□□

Unsupervised Learning — 78% · C

Clustering (K-Means, GMM)	79%	C	■■■■■■■■■■■□□□□
Dimensionality Reduction (PCA)	78%	C	■■■■■■■■■■■□□□□

CS203 · Software Engineering — 90% Grade: A

System Design — 83% · B

Design Patterns	89%	B	■■■■■■■■■■■■■□□
API & Architecture	77%	C	■■■■■■■■■■■□□□□

Dev Practices — 76% · C

Testing & TDD	76%	C	■■■■■■■■■■■□□□□
Version Control & CI/CD	75%	C	■■■■■■■■■■■□□□□

CS204 · Databases — 86% Grade: B

Relational DBs — 88% · B

SQL & Query Optimization	80%	B	■■■■■■■■■■■■□□□
Normalization & Schemas	95%	A	■■■■■■■■■■■■■■□

NoSQL & Big Data — 80% · B

Document & Graph Stores	80%	B	■■■■■■■■■■■■□□□
Data Pipelines	79%	C	■■■■■■■■■■■□□□□

Concept Dependency Map

How topics connect across modules — prerequisites and conceptual overlaps.

From Topic	Module	→	To Topic	Module	Type	Why It Connects
Eigenvalues & Eigenvectors	*Linear Algebra*	→	Dimensionality Reduction (PCA)	*Machine Learning*	**Prerequisite**	*PCA is computed via eigendecomposition of the covariance matrix*
Matrix Decomposition	*Linear Algebra*	→	Linear & Logistic Regression	*Machine Learning*	**Prerequisite**	*Regression solutions rely on matrix inversion & least-squares methods*
Bayes Theorem	*Probability & Statistics*	→	Decision Trees & Ensembles	*Machine Learning*	**Prerequisite**	*Probabilistic splitting criteria (e.g. Naive Bayes) use Bayes directly*
Max Likelihood Estimation	*Probability & Statistics*	→	Linear & Logistic Regression	*Machine Learning*	**Prerequisite**	*MLE is the theoretical foundation for training supervised models*
Distributions & Expectation	*Probability & Statistics*	→	Clustering (K-Means, GMM)	*Machine Learning*	**Prerequisite**	*Gaussian Mixture Models are defined entirely by probability distributions*
Dynamic Programming	*Algorithms & Data Structures*	→	Decision Trees & Ensembles	*Machine Learning*	**Overlap**	*Tree pruning and optimal splits share DP-style subproblem decomposition*
Shortest Paths	*Algorithms & Data Structures*	→	Document & Graph Stores	*Databases*	**Overlap**	*Graph DB engines use Dijkstra & BFS for relationship traversal*
API & Architecture	*Software Engineering*	→	Data Pipelines	*Databases*	**Overlap**	*ETL pipeline design mirrors service-oriented architecture patterns*
SQL & Query Optimization	*Databases*	→	Divide & Conquer	*Algorithms & Data Structures*	**Overlap**	*Query optimizers apply cost-based divide-and-conquer planning*

Academic Advisor: Prof. Sarah Lin ___________ Registrar: _______________ Student Signature: _______________